PROJECT DIARY REPORT

END SEMESTER EVALUATION (PHASE – II)

Intelligent Pesticide Sprinkling System
Determined by the Infection Level of Plants

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Total Duration	9 Weeks (Weeks 1–4 evaluated at Mid-Semester)
Coverage of this Report	Weeks 5–9 (Final 60% of project work)

The first four weeks, presented during the mid-semester evaluation, covered the initial 40% of the project, comprising literature study, dataset preparation, model finalisation, base circuit design, and preliminary subsystem testing. The remaining 60% of the work, reported below, focuses on hardware enclosure assembly, integration of the pump/sprayer mechanism, completion of the web dashboard, end-to-end workflow testing, and final prototype readiness for demonstration.

Week 5 & Week 6 (Combined)

Objectives Planned	To assemble all hardware components inside a compact enclosure and integrate the sprayer mechanism with the control system.
Work Executed During the Week	Mounted and wired the components within the enclosure and integrated the sprayer mechanism, verifying spraying response against infection severity.
Discussion and Feedback from Supervisor	The supervisor appreciated the neat enclosure layout and advised proper wire management and severity-based calibration.
Issues and Challenges Identified	Loose connections caused intermittent operation and the nozzle initially produced an uneven spray pattern.
Action Plan for Next Week	To complete the web dashboard and begin integrated testing of the full image-capture-to-spraying workflow.

Week 7, Week 8 & Week 9 (Combined)

Objectives Planned	To complete the web dashboard and prepare a tested final prototype for demonstration and evaluation.
Work Executed During the Week	Completed the web dashboard and integrated the full image-capture-to-severity-based-spraying workflow, tested on healthy and infected leaf samples.
Discussion and Feedback from Supervisor	The supervisor confirmed the workflow met the objectives and recommended minor UI refinements and additional test runs.
Issues and Challenges Identified	Varying lighting affected severity consistency and dashboard readings needed synchronisation with the spraying action.
Action Plan for Next Week	To carry out final calibration, documentation, and demonstration of the completed prototype for the end-semester evaluation.

Summary: By the end of Week 9, the prototype was fully assembled within a compact hardware enclosure, the sprayer mechanism was integrated with the control system, the web dashboard was operational, and the complete image-capture-to-spraying workflow was tested and ready for demonstration — completing the remaining 60% of the project beyond the 40% achieved at the mid-semester stage.

Signature of Students

Signature of Supervisor

Dr. Karan Veer